Audit Trail — Case 18: Traffic control around a work area near an intersection, one lane closed

Field	Value
Plan status	3 changes · 4 needs attention · 12 checked · 4 pending · reference
MUTCD typical application	TA-22
CDOT standard sheet	S-630-1
Case	Case 18: Traffic control around a work area near an intersection, one lane closed
Taper length	280 ft (L (full merging taper))
Buffer space	305 ft
Device spacing — taper	40 ft
Device spacing — tangent	80 ft
Crew steps	16

Taper Length

Speed 40 mph < 45 mph threshold -> using L = W x S^2 / 60

L = 10.5 x 40^2 / 60 = 280 ft

Required: L = 280 ft (full taper for lane closure)

Field	Value
Closure type	lane
Offset (lane width)	10.5 ft
Required taper (L (full merging taper))	280 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-3. Lane closures use the full merging taper length L.

CDOT reference: CDOT S-630-1 Case 18 (traffic control around a work area near an intersection, one lane closed; Sheet 10)

Buffer Space

MUTCD Table 6B-2: 305 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

40 mph -> 40 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

40 mph -> 80 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

L = 280 ft / 40 ft max spacing -> 8 drums at 40.0 ft intervals

2000 ft / 80 ft max spacing -> 26 cones at 80.0 ft intervals

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 31 tangent + 2 downstream = 33 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 40 mph, road_type = 'urban_low' -> Table 6B-1 category: urban_low

A = 100 ft, B = 100 ft, C = 100 ft

Position	Code	Station (ft)	Distance from Taper (ft)
Advance (S1-1)	S1-1	3,412	800 upstream
C (furthest)	W20-1	2,912	300 upstream
1/2 C inside the outermost sign (Case 18)	R2-10	2,862	250 upstream
B (middle)	W20-5R	2,812	200 upstream
A (nearest)	W4-2R	2,712	100 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: False. Signs placed: 1 left, 12 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 2,912 ft. Required: 2. Placed: 2.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 40 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: True

CDOT Case Reference

Case 18: Traffic control around a work area near an intersection, one lane closed

This scenario follows CDOT Standard Plan S-630-1, Case 18 (Sheet 10) — traffic control around a work area near an intersection, one lane closed — applied to an undivided highway with single-side mainline signing (both-sides posting applies to divided highways, multi-lane ramps, and one-way streets per CDOT S-630-1 Sheet 2 General Note 8), plus a cross-street advance-warning set on each approach leg per MUTCD §6N.12 and the Sheet 10 advance-signing key.

Three disclosed departures from the plate: (1) the Case 18 plate typifies corner-quadrant work with a full cross-street closure train; this plan confines the work space to the mainline and places cross-street advance signing only — corner-quadrant support is tracked at https://github.com/rtmakatura/conestruct/issues/128. (2) The opposing mainline direction is not signed (single-side undivided convention; the plate signs both directions). (3) The plate typifies rural sign placement; urban applications require block-based placement per Sheet 10 Note 1. The plate's Type III corner barricade (work lasting more than 3 days) encloses the corner work area and is not placed, per departure (1).

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

Geometry Validation

Work zone 2000 ft vs required taper 280 ft (L (full merging taper)) + buffer 305 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

Corridor check not run (no site coordinates supplied).

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)	No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure.
MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)	No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure.
MUTCD §6C.02 — pedestrian considerations in work zones	Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.
MUTCD §7B.08 — school zone signing in proximity to work zones	Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 3,412 ft, both sides).

Cross-Street Approaches

Intersection on the far side of the work zone (centerline crossing at mainline station 2200 ft).

Each cross-street leg receives its own advance set (W20-1, R2-10, R2-11), computed from that leg's own speed and road type; stations are measured along the leg from the intersection curb line, increasing away from the intersection.

Approach 'north_leg' — 40 mph, urban_low — SIGNALIZED (signal operation review required). Sheet 10 advance-signing key: A = 100 ft at 40 mph (urban_low).

Code	Station (ft from curb line)	Placement rule
W20-1	200	2 x A = 200 ft, approach side
R2-10	100	A = 100 ft, approach side
R2-11	100	100 ft, departure side (Case 18)

Approach 'south_leg' — 35 mph, urban_low — SIGNALIZED (signal operation review required). Sheet 10 advance-signing key: A = 100 ft at 35 mph (urban_low).

Code	Station (ft from curb line)	Placement rule
W20-1	200	2 x A = 200 ft, approach side
R2-10	100	A = 100 ft, approach side
R2-11	100	100 ft, departure side (Case 18)

Source: MUTCD §6N.12; CDOT S-630-1 Sheet 10, Cases 18/19 (advance-signing key)

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

3 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: https://github.com/rtmakatura/conestruct/issues/117

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (https://github.com/rtmakatura/conestruct/issues/117)
- [interchange_layout_not_generated] An adjacent interchange is flagged, but Conestruct does not generate the per-ramp layout and adds no devices for it. Ramp-specific control design and the ramp-access and coordination review (MUTCD §6N.16, 11th Ed. p. 851) remain with the traffic control supervisor. Tracked at issue #117. (https://github.com/rtmakatura/conestruct/issues/117)
- [signal_operation_review_required] Signalized approaches (north_leg, south_leg): Conestruct places cross-street advance signing only and does not evaluate traffic-signal operation. The signal-operation review — phasing, timing, and signal-head visibility per MUTCD §6N.12 (items 04 and 05; 11th Ed. p. 848) and MUTCD Part 4 — remains with the traffic control supervisor and the operating agency. (https://github.com/rtmakatura/conestruct/issues/117)